FINANCE COURSE

MAJOR PROJECT

EXCHANGE RATE VOLATILITY AND ITS ASYMMETRIC IMPACT ON EXPORT PERFORMANCE

An India-focused study based on published research and secondary information

Submitted in partial fulfilment of the requirements of the Finance Course

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PAGE 02

INDEX

Major Project - Exchange Rate Volatility and Export Performance

S. No.	Particulars	Page
1	Front Page	1
2	Index	2
3	Acknowledgement	3
4	Executive Summary	4
5	Introduction	5
6	Review of Literature - General studies	6
7	Review of Literature - Indian firm and product studies	7
8	Review of Literature - Trade partners and recent studies	8
9	Research Gap and Problem Statement	9
10	Research Objectives, Questions, Scope and Significance	10
11	Research Methodology, Sampling and Data Collection	11
12	Data Analysis and Interpretation - Export context	12
13	Data Analysis and Interpretation - How volatility works	13
14	Data Analysis and Interpretation - Research evidence	14
15	Sector-wise Interpretation and Discussion	15
16	Major Findings	16
17	Limitations of the Study	17
18	Suggestions / Recommendations	18
19	Conclusion	19
20	Bibliography / References	20

Note: This project is a secondary-research study. Page numbers refer to this prepared document. The page sequence follows the Major Project instructions shared for the Finance course.

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I extend my thanks to the authors and researchers whose published studies have helped me understand how exchange-rate changes influence Indian exporters, industries and trade partners. I am especially thankful to the researchers who examined Indian export data and explained the short-run and long-run effects of exchange-rate uncertainty.

I also acknowledge the Ministry of Commerce and Industry, the Government of India, the Reserve Bank of India and the World Trade Organization for making useful economic and trade information available to students and the public. These sources helped me connect academic research with the Indian export environment.

I am grateful to my family, friends and everyone who supported and encouraged me while completing this project. Their help and motivation made it possible for me to organise the materials, read the research papers and complete the work successfully.

Finally, I thank all those who directly or indirectly contributed to the successful completion of this project. I have prepared this study using secondary sources and have tried to present the information in a clear and understandable manner.

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PAGE 04

EXECUTIVE SUMMARY

Exchange rates are important to international trade because exporters receive payments in foreign currencies while paying many production and operating costs in domestic currency. When the value of the rupee changes against currencies such as the US dollar, the rupee value of export receipts, the foreign-currency price of goods and the cost of imported inputs may also change. When exchange rates move sharply and unpredictably, exporters face difficulty in preparing quotations, fixing prices, planning cash flows and deciding when to enter export contracts.

This project studies the relationship between exchange-rate volatility and India's export performance, with special attention to asymmetric effects. An asymmetric effect means that an appreciation and a depreciation, or positive and negative exchange-rate shocks, do not necessarily produce equal effects. Their impact may differ in size, timing, direction and duration.

The project is based on secondary information from published research papers and official trade publications. Sharma's commodity-level study of around 100 Indian merchandise products reports negative short- and long-run effects of exchange-rate volatility on exports, with nominal exchange-rate uncertainty having a particularly strong long-run effect [1]. Other Indian studies report that effects differ across firms, sectors and trading partners [2]-[5].

Official Government of India releases show that total exports (merchandise and services) reached a reported record of US$863.1 billion in FY 2025-26, including merchandise exports of US$441.8 billion and services exports of US$421.3 billion [6]. This growth is important context, but it does not by itself prove that exchange-rate movements caused the increase.

The project concludes that stable and predictable exchange-rate conditions, affordable hedging facilities, better risk awareness and stronger export competitiveness can help exporters manage currency uncertainty. It also explains that depreciation alone cannot guarantee export growth because imported inputs, global demand, productivity and sectoral conditions matter.

PAGE 05

INTRODUCTION

Meaning and importance of the topic

1.1 Meaning of Exchange Rate

An exchange rate is the price of one country's currency expressed in another country's currency. For example, the rupee-US dollar exchange rate indicates how many Indian rupees are required to obtain one US dollar. An increase in the rupees required for one dollar generally represents a depreciation of the rupee against the dollar. A decrease generally represents an appreciation, subject to the quotation convention used.

1.2 Meaning of Exchange-Rate Volatility

Exchange-rate volatility refers to the degree and frequency of variation in the exchange rate over a period. A currency may appreciate or depreciate gradually, or it may move sharply within a short period. Volatility is about uncertainty and the size of movements, not merely whether the currency is strong or weak. Two periods can have the same average exchange rate but very different levels of volatility.

1.3 Meaning of Export Performance

Export performance refers to how well a country or business sells goods and services in overseas markets. It may be measured through export value, export volume, market share, growth rate, number of products exported, number of destination markets and profitability. Higher export value does not always mean higher profit, because costs and exchange-rate conversion also matter.

1.4 Meaning of Asymmetric Impact

An asymmetric impact occurs when positive and negative exchange-rate changes have different effects. For example, depreciation may improve the rupee value of export receipts, but the benefit may be limited when exporters use imported raw materials or have foreign-currency debt. Appreciation may reduce price competitiveness, but it may also lower the cost of imported machinery and inputs. Therefore, the two directions need not cancel each other out.

PAGE 06

REVIEW OF LITERATURE

International evidence and general research

2.1 Auboin and Ruta (World Trade Organization, 2011)

Auboin and Ruta reviewed the economic literature on exchange rates and international trade. Their review considered exchange-rate volatility and currency misalignment. They concluded that volatility has, on average, a negative but not very large effect on trade flows. The strength of the effect depends on factors such as access to hedging instruments, production structure and the degree of integration between countries [7].

Relevance to this project: The review explains why the impact of volatility is not identical for every exporter. A large firm with currency-risk management facilities may respond differently from a small firm that cannot afford hedging.

2.2 Dekle and Ryoo (2007)

Dekle and Ryoo examined exchange-rate fluctuations, financing constraints, hedging and export volumes using firm-level data for Japanese industries. They found that export volumes at the firm level were significantly affected by exchange-rate fluctuations. Their work highlights that financial constraints and access to hedging can influence how firms respond to currency risk [8].

Relevance to this project: The research supports the need to study exporters at the firm level, not only national trade totals. Aggregate exports may hide differences between firms with different financial strength and risk-management practices.

2.3 General Interpretation

The reviewed literature suggests that exchange-rate volatility does not operate in isolation. Its effect depends on the exporter’s cost structure, currency of invoicing, access to financial instruments, market competition and the length of the contract. This provides the background for examining India-specific studies.

REVIEW OF LITERATURE

Indian firm-level and commodity-level studies

2.4 Cheung and Sengupta (2013)

Cheung and Sengupta studied the impact of real effective exchange-rate movements on the export shares of Indian non-financial firms during 2000-2010. Their analysis found that currency appreciation and exchange-rate volatility had a negative effect on firms' export shares. The effect of appreciation was stronger than the effect of depreciation. They also reported that smaller exporters and service-exporting firms were more affected in their sample [2].

Relevance: This study directly supports the asymmetric perspective. It indicates that appreciation and depreciation may not have equal effects and that the size and business type of an exporter matter.

2.5 Sharma (2019; published in 2020 issue)

Chandan Sharma examined exchange-rate volatility and exports from India using monthly panel data for around 100 merchandise products covering December 2012 to November 2017. The study measured volatility using nominal and real exchange-rate concepts and applied a GARCH-based volatility measure with panel estimators. It reported negative short-run and long-run effects on exports. In the long run, both real and nominal exchange-rate volatility mattered, while nominal exchange-rate uncertainty had the stronger effect. In the short run, nominal uncertainty was the main adverse factor reported [1].

Relevance: This is closely related to the present topic because it uses Indian commodity-level export data and distinguishes nominal from real exchange-rate uncertainty.

2.6 Interpretation

Together, these studies show that currency uncertainty may influence the export shares of individual firms and the export performance of different product groups. They also indicate that the time horizon matters: short-term and long-term effects can differ.

PAGE 08

REVIEW OF LITERATURE

Trade partners, sectoral effects and recent studies

2.7 Bahmani-Oskooee and Sahab (2021)

Bahmani-Oskooee and Sahab examined the effects of exchange-rate volatility on India's exports to and imports from its 14 largest trading partners. They found short-run asymmetric effects in almost all partner cases, with long-run asymmetric effects in nearly half of the cases. The results were partner-specific, showing that the response of trade to currency uncertainty differs across destination countries [3].

2.8 Sohrabji (2024)

Sohrabji analysed exchange-rate effects on India's trade using annual data from 1994 to 2022. The study considered income, exchange rates, exchange-rate variability, sectoral differences and a structural break. It reported that exports responded to the real effective exchange rate in the short run but not in the long run, while imports showed a different time pattern. It also found asymmetric exchange-rate effects in high-volume and primary sectors [4].

2.9 Ganaia and Khan (2025)

Ganaia and Khan studied the J-curve phenomenon and export competitiveness in India-US trade over 1980-2023 using a nonlinear ARDL approach. Their results show that short-run and long-run responses differ. The study highlights the importance of manufacturing competitiveness and industrial production capacity alongside the exchange rate [5].

2.10 Overall Review

The literature does not support a single rule that every depreciation increases exports or every appreciation reduces them by the same amount. Instead, effects vary by product, firm, service or merchandise sector, trading partner and time horizon. This variation is the main reason for studying asymmetric impact.

PAGE 09

RESEARCH GAP AND PROBLEM STATEMENT

3.1 Research Gap

Many studies examine the relationship between exchange rates and exports, but their findings are not identical. Some studies focus on national trade totals, while others study firms, product groups or individual trading partners. Some examine the volatility of exchange rates, while others focus on appreciation and depreciation. The different data periods, measures and methods make direct comparison difficult.

For India, the literature shows evidence of negative effects from volatility, but also shows that the effects vary by product, firm size, service or merchandise activity, trading partner and short-run or long-run period. A simple statement that a weaker rupee always benefits exporters is therefore incomplete.

This project addresses the learning gap by bringing together published Indian studies and official export information in one simple review. It explains the difference between exchange-rate movement and exchange-rate volatility, and it highlights why appreciation and depreciation can affect exporters differently.

3.2 Problem Statement

Indian exporters operate in a market where receipts, input costs and competition may be influenced by foreign exchange movements. Unpredictable changes can make it difficult to estimate future revenue and profit. The central problem considered in this study is how exchange-rate volatility affects export performance and why the impact may be different when the currency appreciates compared with when it depreciates.

3.3 Need for the Study

- To improve understanding of currency risk in international business.
- To explain why exporters may experience different effects from similar exchange-rate movements.
- To connect published findings with India’s current export environment.
- To identify practical steps that can help exporters reduce avoidable foreign-exchange risk.

PAGE 10

RESEARCH OBJECTIVES, QUESTIONS AND SCOPE

4.1 Primary Objective

To study the impact of exchange-rate volatility on export performance, with special reference to the asymmetric effects of appreciation and depreciation in the Indian context.

4.2 Secondary Objectives

1. To understand the meaning and causes of exchange-rate volatility.
2. To explain how appreciation and depreciation can influence export prices, receipts and input costs.
3. To review selected research studies on Indian exports and exchange-rate risk.
4. To identify differences in short-run and long-run effects.
5. To examine how responses differ among firms, products, sectors and trading partners.
6. To understand the role of hedging and financial risk management.
7. To provide practical suggestions for exporters and policymakers.

4.3 Research Questions

- Does exchange-rate volatility create uncertainty for Indian exporters?
- Do appreciation and depreciation affect exports in the same way?
- Do effects differ across products, sectors and trading partners?
- Can hedging and better financial planning reduce exposure?

4.4 Scope and Significance

The project focuses on India and uses published studies that cover different periods, firms, commodities and trading partners. It considers merchandise and services exports where the research evidence permits. The study is significant for students, exporters, finance professionals and policymakers because it explains why currency stability, export competitiveness and risk management should be considered together.

PAGE 11

RESEARCH METHODOLOGY

Research design, sampling procedure and data collection

5.1 Research Design

The project follows a descriptive and analytical research design based on secondary data. It reviews published research and official trade information to explain the relationship between exchange-rate volatility and export performance. No new econometric model has been estimated for this student project.

5.2 Nature and Sources of Data

The study uses secondary information from peer-reviewed journal articles, a World Trade Organization working paper and Government of India trade releases. The major research sources examine Indian firms, approximately 100 merchandise products, India's 14 major trading partners and India-US trade. Official trade releases provide context for India's export scale [1]-[7].

5.3 Sampling Procedure and Design

Since the project is based on secondary research, no individual respondents were selected and no questionnaire sample was created. A purposive selection method was used to select published studies that directly examine exchange-rate movements, volatility, asymmetry or Indian export performance. Preference was given to research that clearly stated its country coverage, period, method and findings.

5.4 Period Covered

The reviewed research covers different periods, including 2000-2010, 2012-2017, 1994-2022 and 1980-2023. These are the periods used by the original authors and should not be interpreted as one continuous dataset created by this project.

5.5 Tools and Method of Analysis

- Descriptive comparison of published research findings.
- Thematic analysis of depreciation, appreciation, volatility and export outcomes.
- Comparison of short-run and long-run effects.
- Use of official export statistics for background context.

Important: The project does not claim to have conducted interviews, Google Forms, a survey, or original statistical tests. All empirical results are attributed to the original researchers.

PAGE 12

DATA ANALYSIS AND INTERPRETATION

India’s export context

6.1 Recent Export Position

India’s export sector is large and includes both merchandise and services. A Ministry of Commerce and Industry release dated 28 July 2026 reported total exports of US$863.1 billion for FY 2025-26, made up of US$441.8 billion in merchandise exports and US$421.3 billion in services exports [6]. This is an important background indicator of the scale of India’s international trade.

Export category	FY 2025-26 reported value	Share of reported total (approx.)
Merchandise exports	US$441.8 billion	51.2%
Services exports	US$421.3 billion	48.8%
Total exports	US$863.1 billion	100.0%

The approximate shares above are calculated from the rounded figures in the Government release. They describe the composition of the reported total, not the effect of exchange-rate movements.

6.2 Interpretation

The figures show that services exports contribute a substantial part of India’s export earnings, alongside merchandise exports. Therefore, exchange-rate risk management is relevant not only to manufacturers and traders of physical goods but also to service providers who invoice overseas clients and receive foreign-currency payments.

However, national export growth cannot be used as proof that depreciation improved exports. Export values may rise because of higher foreign demand, greater production, improved product quality, new trade agreements, global commodity prices or changes in service delivery. A separate econometric design is required to isolate the effect of exchange-rate volatility.

6.3 Data Note

The Ministry published an earlier FY 2025-26 estimate in April 2026 and a later record figure in July 2026. This project uses the later July release for the current context and does not combine values from different release vintages in one growth calculation [6].

PAGE 13

DATA ANALYSIS AND INTERPRETATION

How exchange-rate volatility affects exporters

6.4 Price Competitiveness Channel

When the rupee depreciates, an Indian product priced in rupees may become cheaper for a foreign buyer if the exporter passes the exchange-rate benefit into the foreign-currency price. This may support demand. But the exporter may choose to keep the foreign-currency price unchanged and receive more rupees instead. The actual result depends on pricing decisions and competition.

6.5 Cost and Profit Channel

Many exporters use imported fuel, components, machinery, chemicals or technology. If the rupee depreciates, these inputs may become more expensive in rupee terms. The increase in input costs can reduce or even offset the benefit from higher rupee export receipts. A firm with large imported-input requirements may respond differently from a firm using mostly domestic inputs.

6.6 Uncertainty Channel

Volatility makes future exchange rates difficult to predict. Exporters may hesitate to quote long-term prices, accept orders with delayed payment or make new investments. A business may also face a difference between the exchange rate assumed when a contract is signed and the rate available when payment is received.

6.7 Asymmetric Response

Type of movement	Possible effect	Why it may be different
Depreciation	Can improve export price competitiveness and rupee receipts	Imported inputs, foreign-currency debt and buyer contracts may reduce the benefit
Appreciation	Can lower rupee receipts and make exports less price-competitive	Imported inputs may become cheaper; productivity and pricing power may offset effects
Volatility in either direction	Can make cash flows and pricing uncertain	Hedging access, firm size and contract duration affect exposure

Interpretation: The table describes economic channels, not guaranteed outcomes. The net effect must be assessed using an exporter’s own costs, currencies, contracts and destination markets.

PAGE 14

DATA ANALYSIS AND INTERPRETATION

Comparison of published empirical findings

Study	Coverage / method	Main finding relevant to this topic
Cheung & Sengupta (2013) [2]	Indian non-financial firms, 2000-2010	Appreciation and volatility reduced export shares; appreciation effect was stronger than depreciation. Smaller exporters and service firms were more affected in the sample.
Sharma (2019/2020) [1]	Around 100 Indian merchandise products, Dec 2012-Nov 2017; GARCH-based volatility and panel estimators	Volatility had negative short- and long-run effects; nominal exchange-rate uncertainty was especially important in the long run.
Bahmani-Oskooee & Sahab (2021) [3]	India's trade with 14 major partners; asymmetric trade models	Short-run asymmetry appeared for almost all partners; long-run asymmetry for nearly half. Effects were partner-specific.
Sohrabji (2024) [4]	India annual data, 1994-2022; dynamic fixed effects	Exports responded to REER in the short run but not the long run; exchange-rate effects differed across sectors.
Ganaia & Khan (2025) [5]	India-US trade, 1980-2023; NARDL and related tools	Short- and long-run trade-balance responses differed; manufacturing competitiveness and productive capacity were important.

6.8 Interpretation of the Evidence

The studies broadly support the view that currency risk can matter for export outcomes, but the strength and direction of the effect are not uniform. Sharma reports an adverse effect of volatility in Indian merchandise exports. Cheung and Sengupta identify an asymmetric firm-level response. The partner-level study by Bahmani-Oskooee and Sahab shows that the destination country matters. Sohrabji also reports sectoral variation. These differences demonstrate why national totals alone are insufficient for understanding export performance.

SECTOR-WISE INTERPRETATION AND DISCUSSION

6.9 Merchandise Exporters

Merchandise exporters may be exposed to currency risk through raw materials, imported components, freight, insurance and foreign-currency payments. A depreciation can support competitiveness when domestic production costs are largely in rupees. The same depreciation can increase costs when imported content is high. Therefore, the net outcome is product-specific.

6.10 Services Exporters

Service exporters such as information technology and business-service providers may receive foreign-currency payments while paying salaries and other expenses in rupees. This may create a different pattern of currency exposure from manufacturers. Cheung and Sengupta report that service firms in their sample were more affected by exchange-rate fluctuations than goods exporters [2]. This should be understood as a finding from that study’s sample, not a universal rule for every service company.

6.11 Small and Large Exporters

Large firms may have treasury teams, multiple currency accounts, diversified markets and access to forward contracts. Small exporters may have less bargaining power, fewer financial resources and limited access to professional risk management. The WTO literature review also notes that the production structure and availability of hedging instruments influence how volatility affects trade [7].

6.12 Trading Partner Differences

A currency shock may affect exports differently depending on the destination market, the currencies used for invoicing, the competitiveness of local suppliers, and demand conditions in that country. The evidence from India’s 14 major partners indicates that asymmetry is partner-specific [3]. Exporters should therefore assess exposure by market rather than rely only on one rupee-dollar indicator.

6.13 Overall Discussion

The evidence suggests that exchange-rate management is one part of export competitiveness. Productivity, quality, delivery time, logistics, trade finance, innovation and market access remain important. Currency depreciation cannot compensate indefinitely for weak productivity or high production costs.

PAGE 16

MAJOR FINDINGS

8. Exchange-rate volatility creates uncertainty in export pricing, receipts, costs and cash-flow planning.
9. The effect of volatility is not identical for every exporter; it depends on firm size, imported-input dependence, financial strength and access to hedging.
10. Indian empirical research reports negative short-run and long-run export effects from exchange-rate volatility in the commodity-level study by Sharma [1].
11. Appreciation and depreciation may have unequal effects. Cheung and Sengupta found a stronger negative effect from appreciation than depreciation for the firms studied [2].
12. The impact differs across destination markets. Research on India's 14 major trading partners found partner-specific short-run and long-run asymmetries [3].
13. Sectoral effects also vary. Sohrabji found differences across high-volume and primary sectors and different time responses for exports and imports [4].
14. The short-run and long-run effects should be analysed separately. A currency movement may have an immediate pricing effect and a different later production or trade-volume effect.
15. Depreciation may improve the rupee value of export receipts but can increase the cost of imported raw materials and components.
16. Access to hedging instruments can reduce some short-term currency risk, but it does not remove all business and market uncertainty [1], [7].
17. India's record export value in FY 2025-26 shows the importance of the export sector, but aggregate growth cannot by itself establish the causal impact of exchange-rate volatility [6].
18. Export competitiveness depends on more than the exchange rate. Productivity, quality, technology, infrastructure, logistics and global demand also matter.

PAGE 17

LIMITATIONS OF THE STUDY

The project has the following limitations, which should be considered while interpreting its findings:

- The study is based on secondary information and does not include a primary survey, interviews or questionnaire responses.
- The project does not estimate a new econometric model or independently calculate exchange-rate volatility. It summarises results reported by the original researchers.
- The research papers use different periods, variables, countries, products, sectors and econometric methods. Their results are not directly interchangeable.
- Some studies examine exchange-rate movements or the real effective exchange rate, while others examine volatility or asymmetric shocks. These concepts are related but not identical.
- The project uses official export statistics for context. Aggregate export values are influenced by many variables and cannot isolate the effect of exchange rates.
- The latest export statistics may be revised by the Government after publication. The figures quoted here reflect the cited release date.
- The findings of one study may not apply to all Indian exporters, because firms differ in size, imported-input share, pricing power and risk-management capacity.
- Exchange-rate effects may vary across destination countries and contract periods. The project cannot provide a forecast for a particular firm or currency pair.
- The project does not provide investment, trading or foreign-exchange dealing advice. Businesses should consult qualified professionals before entering hedging contracts.

8.1 Scope for Further Study

Future research may use updated monthly firm-level or product-level export data, a clearly defined exchange-rate volatility measure, imported-input exposure, destination-market demand and separate short-run and long-run models. A primary survey of small exporters could also help understand practical difficulties in using hedging instruments.

PAGE 18

SUGGESTIONS / RECOMMENDATIONS

9.1 Suggestions and Recommendations

- Improve awareness of foreign-exchange risk: Exporters should understand how currency movements affect contract prices, receipts, costs and margins.
- Use suitable hedging carefully: Exporters may evaluate forward contracts and other permitted risk-management instruments with authorised banks and qualified advisers.
- Match receipts and payments: Where practical, firms can align foreign-currency inflows and outflows to reduce the need to convert currencies repeatedly.
- Prepare sensitivity budgets: Businesses should assess how different exchange-rate scenarios may affect revenue, imported input costs and profit margins.
- Support small exporters: Banks, export-promotion bodies and industry associations should provide simple training and accessible information about currency risk.
- Diversify products and markets: Dependence on one product, one buyer or one destination market can increase vulnerability to shocks.
- Improve productivity and quality: Technology, skills, delivery reliability and cost control should be strengthened so that competitiveness does not depend only on currency movements.
- Use reliable information: Exporters should monitor official trade information, central-bank publications and destination-market conditions before making business decisions.

PAGE 19

CONCLUSION

10.1 Conclusion

Exchange-rate volatility is an important financial risk for exporters because it affects the value of foreign-currency receipts, export prices, imported input costs and business planning. The reviewed Indian studies show that these effects are not uniform. Appreciation and depreciation may have unequal effects, and the result can change by firm size, sector, product, trading partner and time period.

The evidence does not support the idea that depreciation alone guarantees export growth. Export performance depends on a wider group of factors, including demand, productivity, quality, technology, logistics and access to finance. Therefore, exporters should combine currency-risk management with long-term improvements in business competitiveness. Predictable exchange-rate conditions and wider access to risk-management knowledge can support a stronger and more resilient export sector in India.

PAGE 20

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Note: References [1]-[5] are research studies; [6] and [9] provide official Indian trade context; [7] and [8] provide broader research background. URLs were checked during preparation. Accessed 6 November 2026.